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Epidemiologic Trends in Hand Injuries in the National Football League

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ABSTRACT

Objectives: In American football, hand injuries have been shown to negatively impact performance. The purpose of this study is to characterize the prevalence and severity of hand injuries in National Football League (NFL) players.

Methods: A public online database was utilized to identify hand injuries in NFL players from 2009–2010 to 2019–2020. The primary outcome was to analyze the overall incidence of hand injuries (including wrist, metacarpus, finger, and thumb), injury type by each aforementioned anatomic location, and player position. Injury severity was evaluated based on percentage of injuries in which players returned to play (RTP), number of games missed before RTP, and the percentage of injuries resulting in the player being placed on injured reserve (IR).

Results: Of the 6,127 players included, 847 (13.8%) players sustained a hand injury, of which 24.8%, 34.3%, 17.9%, and 22.9% occurred at the wrist, metacarpus, finger, and thumb, respectively. Of the injured players, 97.4% returned to play following their injury, 14.8% were put on IR, and an average of 1.7 (SD 3.3) games were missed. Quarterbacks were the most likely to sustain hand injuries at all anatomic locations. Wrist injuries were associated with the lowest RTP rate (93.3%), the most players placed on injured reserve (28.6%), and the greatest number of games missed (mean 2.5, SD 4.2).

Conclusion: Hand injuries decreased in prevalence by 65.6% over the 11 NFL seasons evaluated. This trend coincides with the implementation of several safety rules that relate to components of play involving the hands. Quarterbacks experienced the greatest prevalence and severity for all hand injuries. Wrist injuries represent the anatomic location associated with the greatest severity. These findings may be able to inform tailored injury prevention practices by position, and advocate for the further adoption of safety rules to protect players from further injury.

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Keywords: American Football, Football Injuries, Wrist, Finger, Thumb, Metacarpus, Sports Injuries, Injury Incidence

INTRODUCTION

In every play on the field, American football players put their hands at risk, yet there is a limited body of recent research that addresses the nature and pattern of hand injuries. From 2009 to 2019, National Collegiate Athletic Association (NCAA) football players sustained over 2,200 hand or wrist injuries across all player positions[1–3]. The most common injuries in this cohort were thumb sprains, metacarpophalangeal (MCP) joint dislocations, radial/ulnar collateral ligament injuries, proximal interphalangeal joint dislocations, jersey fingers, mallet fingers, boutonniere deformities, metacarpal fractures, avulsion fractures, and scaphoid fractures[1,4]. A study reviewing injuries screened for during the National Football League (NFL) Combine from 1987 to 2000, where NCAA players are evaluated prior to the NFL draft, observed hand injuries comprised 33.5% of all identified injuries[5]. Furthermore, hand injuries were found to negatively affect player performance across multiple positions, with defensive backs recording fewer tackles per game, running backs recording fewer carries/yards per game, and tight ends playing significantly fewer games[5].

Mall et al. found that from 1996 to 2005, the most common hand injuries in the NFL were metacarpal fractures and interphalangeal dislocations[6]. Additionally, they found that linemen, wide receivers, and defensive backs were the positions most likely to experience hand injuries. Notably, non-skill positions (offensive line, defensive line, linebackers, defensive backs) returned to play in a fewer number of days compared to skill positions (quarterbacks, running

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backs, wide receivers), and were less likely to require surgical repair, likely due to the high stress levels that throwing, carrying, and catching have on the hand[7,8]. Similar trends in hand injuries have been reported at the collegiate level. Brophy et al. measured the prevalence and trends of musculoskeletal injuries among NCAA football players from 1987-2000, finding soft-tissue hand and wrist injuries to be second highest incidence per 100 players, behind that for ankle sprains. MCP dislocations and phalanges fractures were reported to have the 10th highest incidence, with the second fastest-growing incidence during the study period[9].

Stylistic and technological advancements among coaches, players, and medical staff, within the past 20 years, may have changed the incidence, distribution, and management of hand injuries.[10] These advancements include new safety regulations that mitigate injury risk, as well as improved injury screening and treatment strategies.[11] One demonstrative example is the "targeting rule", which was implemented in 2018 and penalizes a player who lowers his head to strike another player with his helmet.[12] Adherence to the targeting rule has been found to reduce concussions without increasing the number of lower extremity injuries.[12] Other rule changes to limit the occurrence of dangerous plays have been continually implemented over the past decade but have not been extensively studied.[13] It is important to understand how these systemic changes have resulted in differences in prevalence and severity of hand injuries to help influence orthopaedic care.

Despite the present body of literature showing the negative impact of hand injuries on player performance, there is a paucity of recent research that characterizes the prevalence of hand injuries in football players. Understanding which anatomic locations within the distal upper extremity are most commonly injured among NFL players, as well as the football positions in which players are at increased risk for certain hand injuries, is crucial for the implementation of

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specific and effective injury prevention practices. Therefore, the purpose of this study was to assess the overall prevalence of hand injuries in NFL players based on injury type (wrist, metacarpus, finger, thumb) and player position from the 2009-2010 season through the 2019-2020 season. We hypothesized that the prevalence of hand injuries has decreased over this time period, in light of the recent safety rules implemented to protect players.

METHODS

Data Source

This study was a retrospective review of the publicly available Pro Football Reference online database[14]. Pro Football Reference collects statistics on various parameters from many aspects of the NFL, going as far back as 1960. Each team has a database of injuries reported by year. Any injury listed on a team's injury report regardless of severity is recorded in the Pro Football Reference database. Each injury is classified by the general anatomic location affected. Every athlete is monitored weekly to track the number of games missed and whether the player was able to return to play. This data source has been utilized in several peer-reviewed studies on injury incidence and outcomes among NFL players.[15–20]

Hand injury prevalence for each of the 32 NFL teams from the 2009-2010 to the 2019-2020 seasons was queried. This timeframe was chosen because the Pro Football Reference injury database began collecting and reporting data in 2009, and we did not want to introduce any variability that may have been caused by irregularities during the 2020-2021 season that was played during the COVID-19 pandemic. The database subclassified hand injuries into the following categories: finger, hand, thumb, and wrist. To avoid confusion with overall hand

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injuries and the exclusive category of hand injury, we refer to the exclusive hand category as "metacarpus". Players were excluded from the injured cohort if they had any injury not related to the hand. Players with concomitant injuries to other parts of the body were included, so long as injury report specified the player's injury as "hand", "wrist", "finger", or "thumb" as the reason for any games missed. Players with multiple hand injuries were counted for each hand injury they sustained.

Variables in the Database

Variables observed among all players included BMI (mean and standard deviation (SD)), number of seasons played within the study period, and player position. Player positions included center, defensive back, defensive line, kicker, linebacker, long snapper, offensive line, punter, quarterback, running back, tight end, and wide receiver. Though centers are part of the offensive line, they were separated for this analysis due to their increased level of hand involvement compared to other offensive linemen, as centers directly handle the ball on each play. The team to which each player belonged at the time of injury was noted but not factored into the analysis. Variables observed for those injured included hand injury location, player position, average games missed (mean and SD), percentage of players placed on injured reserve (IR), and percentage of players who returned to play (RTP). IR is a designation for players who have suffered any injury that makes them ineligible to play for the team, allowing the team to bring another player onto the roster to replace the injured player while they recover[21].

Data Analysis

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The prevalence of hand injuries, overall and stratified by anatomic location, were identified for each season. The prevalence and severity of injury were then assessed based on anatomic location. The number of games missed, RTP, and players on IR following these injuries were used as surrogates to determine injury severity. A sub-analysis of the overall prevalence and severity of injury was performed for all hand injuries and for the different anatomical locations. Data analysis was performed using Microsoft Excel.

RESULTS

Player Demographics

In total, 6,127 players were included in this study. There were 178 (2.91%) centers, 1209 (19.73%) defensive backs, 942 (15.37%) defensive linemen, 98 (1.60%) kickers, 858 (14.00%) linebackers, 61 (1.00%) long snappers, 737 (12.03%) offensive linemen, 93 (1.52%) punters, 210 (3.43%) quarterbacks, 583 (9.52%) running backs, 382 (6.23%) tight ends, and 776 (12.67%) wide receivers (**Table 1**). The mean BMI of the cohort was 31.1 (SD 4.6), and the cohort played a mean of 3.64 (2.57) seasons during the study period (**Table 1).**

Hand Injuries by Anatomic Location

During our study period, 726 (11.85%) players sustained at least one hand injury, with 825 total hand injuries. Ten-year trends are shown in **Figure 1**. Total hand injuries decreased in prevalence by 65.6% between 2009 and 2019. Of these total hand injuries, 196 (23.75% of all injuries) were wrist injuries, 288 (34.90% of all injuries) were metacarpus injuries, 150 (18.18% of all injuries) were finger injuries, and 191 (23.15% of all injuries) were thumb injuries. The

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percentage of players who returned to play, mean games missed, and percentage placed on IR following any hand injury were 97.40%, 1.70 (SD 3.31), and 14.77%, respectively. Players who sustained a wrist injury had the lowest percent RTP (93.33%), highest mean games missed (mean = 2.48, SD 1.96), and highest percentage of players placed on IR (28.57%). Metacarpus injuries had the highest return to play (98.97%), but finger injuries resulted in the lowest mean games missed (mean = 0.72, SD 2.04) and percentage on IR (4.61%). **Table 2** details all hand injury outcomes, including general hand injuries and injuries stratified by anatomic location.

All Hand Injuries by Player Position

Quarterbacks were the most likely to sustain any hand injury (26.67%), while kickers were the least likely (0%). Long snappers had the lowest percent RTP (66.70%), highest average games missed (mean = 4.00, SD 5.66), and 100% of injured players placed on IR. Defensive backs had the highest percent RTP (98.82%), whereas the defensive line had the lowest average number of games missed (mean = 0.94, SD 3.32) and lowest percentage on IR (8.76%). **Table 3** characterizes overall hand injuries stratified by player position.

Wrist Injuries by Player Position

Long snappers were the most likely to sustain a wrist injury (6.56%), whereas kickers and punters did not sustain any wrist injuries. Long snappers also had the lowest percent RTP (0.00%) and highest percent of players placed on IR (100%), while tight ends had the highest average games missed (mean = 4.30, SD 5.49). Centers, offensive linemen, and quarterbacks had the highest percent RTP (100.00% for each), whereas defensive linemen missed the fewest average

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number of games (mean = 1.24, SD 4.97) and had the lowest percentage of players placed on IR (15.00%). **Table 4** characterizes wrist injuries stratified by player position.

Metacarpus Injuries by Player Position

Quarterbacks were the most likely to sustain a metacarpus injury (6.67%), and kickers did not sustain any metacarpus injuries. Quarterbacks had the lowest percent RTP (93.33%), whereas long snappers missed the most games on average (mean = 8.00, SD N/A) and had the highest percent of players placed on IR (100%). Centers, defensive linemen, linebackers, long snappers, punters, tight ends, and wide receivers all had 100% RTP for metacarpus injuries. Defensive linemen missed the fewest average number of games (mean = 0.90, SD 1.83), and punters had the lowest percentage of players placed on IR (0.00%), followed by wide receivers (5.00%). **Table 5** characterizes metacarpus injuries stratified by player position.

Finger Injuries by Player Position

Quarterbacks were the most likely to sustain a finger injury (7.62%), while kickers and punters did not sustain any finger injuries. Linebackers (93.33%;) and offensive linemen (94.74%) had the lowest percent RTP, and all other positions with finger injuries had 100% RTP. Defensive backs missed the most games on average (mean = 1.00, SD 1.85), whereas long snappers (mean = 0.00, SD N/A) and running backs (mean = 0.00, SD 0.00) missed the fewest average number of games. Offensive linemen had the highest percent of players placed on IR (10.53%). Centers, defensive linemen, long snappers, running backs, and tight ends had no finger injuries that resulted in placement on IR. **Table 6** characterizes finger injuries stratified by player position.

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Thumb Injuries by Player Position

Quarterbacks were the most likely to sustain a thumb injury (10.95%). Kickers, punters, and long snappers did not sustain any thumb injuries. Centers had the lowest percent RTP (83.33%) and highest percent placed on IR (33.33%), while quarterbacks missed the most games on average (mean = 4.74, SD 3.41). Defensive backs, defensive linemen, linebackers, offensive linemen, tight ends, and wide receivers all had 100% RTP for thumb injuries. Tight ends missed the fewest games on average (mean = 0.00, SD 0.00), and none sustained thumb injuries that resulted in placement on IR. **Table 7** characterizes thumb injuries stratified by player position.

DISCUSSIONS

We found that hand injuries decreased in prevalence by 65.6% over the 11 NFL seasons observed in this study, supporting our hypothesis. Over the study period, there were 825 reported hand injuries, including injuries to the wrist, metacarpus, finger, and thumb. This trend aligns with the decreasing prevalence of other types of injuries in NFL players observed in recent years, including lower extremity injuries and concussions[22,23].

There are several factors that may explain this decreasing prevalence. Since the beginning of our study period in 2009, the NFL has implemented 39 new rules specifically designed to address and improve player safety, of which 23 (59.0%) were implemented in 2013 or later[24]. Of the newly implemented rules, 16 (41.0%) relate to components of play that more directly involve the hands, such as blocking or tackling. 11 of these 16 rules (68.8%) were implemented in 2013 or later. The increase in the overall sheer number and hand-specific player safety-related rules may

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have influenced the prevalence of hand injuries, particularly because the dangerous blocking and tackling plays banned in these new rules were previously responsible for frequent and severe injuries. The majority of these rules going into effect after 2013 may explain why the prevalence started to decrease after its peak in 2012. While this retrospective study cannot prove causation, it is a meaningful finding that the decrease in hand injuries correlates with an increase in safety rules. The implications of these findings may be used to advocate for the further adoption of safety rules to protect players from further injury. Beyond rule changes, is also possible that the decrease in hand injury prevalence could be related to technique optimization among players, however, this is a research subject that has yet to be investigated.

The decreasing prevalence may also be related to injury reporting. Players with mild injuries may be disincentivized to report to medical staff, as reporting may result in being pulled from practices or games. Since many NFL players' salaries are non-guaranteed and partially depend on performance-based incentives, they would be incentivized not to report[25]. This is corroborated by Kerr et al. who identified not wanting to lose playing time, not wanting to let teammates or fans down, feeling pressure from coaches to keep playing, and the societal rewarding of health-related risk taking as the main reasons not to report injuries[26–32]. There is no way to quantify how many injuries go unreported in NFL players, but it is a factor that must be considered when searching for rationale behind trends in injury prevalence.

The present study found that when considering percent RTP, mean games missed, and percent of players played on injured reserve as measures of injury severity, wrist injuries were the most severe while finger injuries were least severe. Football is classified as a collision sport, with players actively using their hands to block or tackle, terminating each play with players being brought forcefully to the ground[2,6,33]. Blocking, tackling, and falling put the wrist at

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risk for dorsiflexion injury upon impact[34]. In fact, it has been previously reported that the most common mechanism of hand and wrist injury in football is player-on-player contact, with tackling and blocking being the actions responsible for the vast majority of these injuries[2,6]. There is a lack of literature explaining mechanisms of injury involving the fingers, namely the second through fifth digits. However, previous studies have suggested that finger injuries are commonly "played through", with athletes most likely underreporting these injuries[35]. Finger dexterity and grip strength are not vital skills for most football positions, which may allow players to feel more comfortable returning to play when compared wrist, metacarpus, or thumb injuries.

Contrary to prior literature, our study showed quarterbacks had the greatest prevalence and severity of all hand injuries based on position. Mall et al. observed in their analysis of NFL athletes from 1996 to 2005 that offensive and defensive linemen were the most likely to sustain a hand injury[6]. However, offensive and defensive linemen make up the majority of each team; thus, they are the most likely to be injured. Our analysis stratifies based on the number of players in each position, observing quarterbacks to have the highest prevalence of hand injuries when compared to all other positions. The quarterback is the position most naturally associated with hand usage on a football team, with injuries most likely associated with being tackled. While there is no literature specific to hand injuries, a study of shoulder injuries in quarterbacks revealed that nearly 70% of injuries occurred while the quarterback was being tackled[36]. Most injuries occurred when the quarterback was tackled as the passer, or sacked (47.3%), followed by tackles occurring when the quarterback acted as the ball carrier (12.6%)[36]. There is no literature to support if these situations induce more forceful tackles from defensive players. However, when a quarterback knows he is about to be sacked, he will often attempt to throw the

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ball away to avoid taking a loss in yardage[37]. Having his arm outstretched or in motion while simultaneously being forcefully tackled may explain the increased prevalence and severity of hand injuries in quarterbacks.

While long snappers had the lowest percent RTP and the greatest percentage of wrist injuries among the position, the generalizability of this data is limited by the small sample size of the position. Only 61 long snappers played during the study period, of which 7 recorded hand injuries. Because long snapper is such a specialized position, with usually only one on each team's roster, a hand injury may compel a team to release the long snapper and rapidly search for a replacement, which could contribute to the lower RTP rate seen among long snappers. However, more data is needed on the effect of hand injuries on long snapper performance and recovery, as this limited sample size cannot fully capture a true trend.

Defensive linemen experienced the lowest prevalence of hand injuries among non-special teams players, which is in concordance with previous studies. Mall et al. reported that the act of blocking was 2.7 times more likely to cause a hand injury than the act of being blocked[6]. Mechanistically, this corroborates the present study's observation of decreased prevalence of hand injuries in defensive linemen.

Together, the results of our study show differences in hand injury prevalence by position, and could be used to facilitate the adoption of position-specific injury prevention protocols to mitigate risk of future hand injuries in the NFL. Future studies should look prospectively at hand injury prevention practices that differ by position to assess whether this could decrease the prevalence of hand injuries in football players.

Our study has several limitations. There are inherent difficulties in working with a public injury database, including verified completeness of the dataset. The Pro Football Reference

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database only classifies injuries by general anatomic location, eliminating the ability to distinguish more specific types of injury, such as fractures, ligament tears, or dislocations. Injuries classified in the database as “other” do not receive an anatomic designation, creating the potential for underrepresentation of hand injuries. Hand injuries would not be reported if players did not seek medical treatment, which could also contribute to underreporting. Some position types are limited in the extent to which their injuries can be analyzed due to small sample size. This study used number of games missed as a surrogate for injury severity, which could be an unreliable measure given that hand injuries may have variable impacts on RTP depending on position. This study did not explore associations that increase risk of hand injuries, as it was outside the scope of this investigation.

CONCLUSIONS

Hand injuries decreased in prevalence by 65.6% from 2009 to 2019, which coincides with an increase in NFL rules implemented for player safety. Of the anatomic locations studied, wrist injuries were associated with the greatest severity. Quarterbacks were the position with the greatest prevalence and severity of all hand injuries. Further research on preventative measures to decrease the prevalence of both hand injuries in quarterbacks, and wrist injuries in all players is needed.

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FIGURE CAPTIONS

Figure 1. Ten-Year Trends in Finger, Metacarpus, Thumb, and Wrist Injuries in the NFL from the 2009 to 2019 Seasons.

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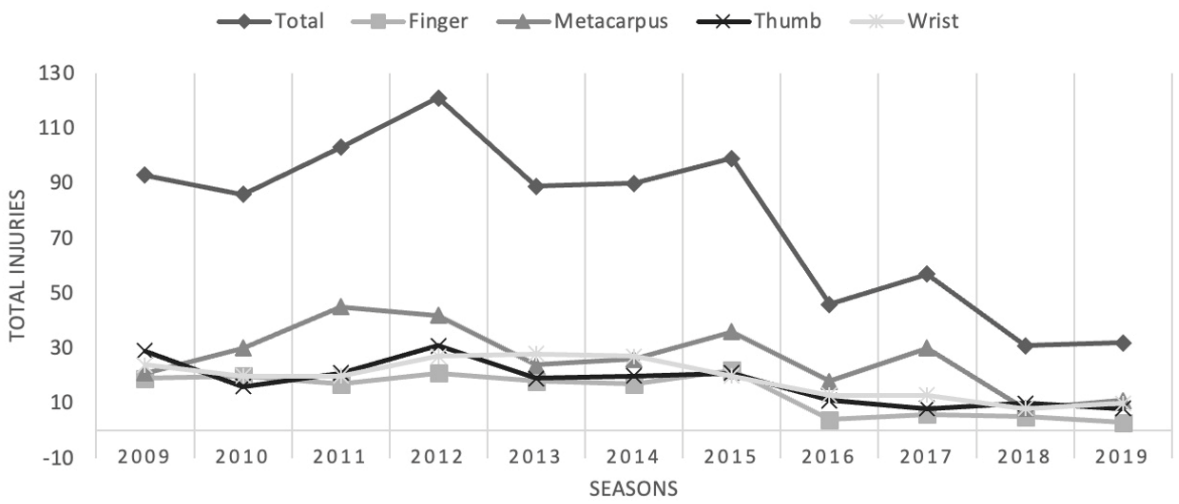

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Table 1. Player Characteristics		
Category	**Number**	**Percent**
Total	6127	-
BMI	Mean = 31.1	(SD = 4.6)
Seasons Played (Years)	Mean = 3.64	(SD = 2.57)
Player Position	-	-
Center	178	2.91%
Defensive Back	1209	19.73%
Defensive Line	942	15.37%
Kicker	98	1.60%
Linebacker	858	14.00%
Long Snapper	61	1.00%
Offensive Line	737	12.03%
Punter	93	1.52%
Quarterback	210	3.43%
Running Back	583	9.52%
Tight End	382	6.23%
Wide Receiver	776	12.67%

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Table 2. Hand Injury Outcomes Stratified by Anatomic Location.				
Position	**Number of All-Hand Injuries**	**% Return to Play**	**Mean Number of Games Missed**	**% Injured Reserve**
All-Hand	825	97.40%	1.70 (SD 3.31)	14.77%
Wrist	196 (23.75%)	93.33%	2.48 (SD 1.96)	28.57%
Metacarpus	288 (34.90%)	98.97%	1.50 (SD 2.63)	12.02%
Finger	150 (18.18%)	98.68%	0.72 (SD 2.04)	4.61%
Thumb	191 (23.15%)	98.45%	1.97 (SD 3.76)	15.46%

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Table 3. Overall Hand Injury Outcomes Stratified by Player Position.				
Position	**Number of Players with Any Hand Injury**	**% Return to Play**	**Number of Games Missed**	**% Injured Reserve**
Total	726 (11.85% of all players)	97.40%	1.70 (SD 3.31)	14.77%
Center	23 (12.92% of all centers)	96.88%	1.32 (SD 1.96)	15.63%
Defensive Back	150 (12.41% of all defensive backs)	98.82%	1.65 (SD 3.32)	13.61%
Defensive Line	97 (10.30% of all defensive linemen)	98.54%	0.94 (SD 2.06)	8.76%
Kicker	0 (0.00% of all kickers)	N/A	N/A	N/A
Linebacker	96 (11.91% of all linebackers)	96.15%	1.97 (SD 3.72)	19.23%
Long Snapper	7 (11.48% of all long snappers)	66.70%	4.00 (SD 5.66)	100.00%
Offensive Line	83 (11.26% of all offensive linemen)	98.30%	2.12 (SD 3.61)	16.47%
Punter	1 (1.08% of all punters)	100.00%	N/A	N/A
Quarterback	56 (26.67% of all quarterbacks)	97.30%	2.26 (SD 3.90)	18.92%
Running Back	68 (11.66% of all running backs)	93.33%	1.66 (SD 3.19)	22.67%
Tight End	39 (5.37% of all tight ends)	97.68%	2.10 (SD 4.10)	16.28%
Wide Receiver	106 (13.66% of all wide receivers)	97.58%	1.73 (SD 3.41)	13.71%

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Table 4. Wrist Injury Outcomes Stratified by Player Position.				
Position	**Number of Players with Wrist Injuries**	**% Return to Play**	**Number of Games Missed**	**% Injured Reserve**
Total	204 (3.33%)	93.33%	2.48 (SD 4.20)	28.57%
Center	1 (0.56%)	100.00%	3.00 (SD 3.00)	40.00%
Defensive Back	37 (3.06%)	97.22%	2.40 (SD 4.63)	22.22%
Defensive Line	33 (3.50%)	95.00%	1.24 (SD 2.48)	15.00%
Kicker	0 (0.00%)	N/A	N/A	N/A
Linebacker	38 (4.43%)	91.89%	2.91 (SD 4.97)	35.14%
Long Snapper	4 (6.56%)	0.00%	N/A	100.00%
Offensive Line	18 (2.44%)	100.00%	2.27 (SD 4.04)	20.00%
Punter	0 (0.00%)	N/A	N/A	N/A
Quarterback	13 (6.19%)	100.00%	3.10 (SD 5.11)	15.38%
Running Back	20 (3.43%)	86.36%	3.58 (SD 4.99)	50.00%
Tight End	10 (2.62%)	90.91%	4.30 (SD 5.49)	45.45%
Wide Receiver	30 (3.87%)	90.00%	2.11 (SD 3.08)	30.00%

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Table 5. Metacarpus Injury Outcomes Stratified by Player Position.				
Position	**Number of Players with Metacarpus Injuries**	**% Return to Play**	**Number of Games Missed**	**% Injured Reserve**
Total	271 (4.42%)	98.97%	1.50 (SD 2.63)	12.02%
Center	11 (6.18%)	100.00%	1.40 (SD 1.72)	6.67%
Defensive Back	64 (5.29%)	98.51%	1.26 (SD 3.32)	10.45%
Defensive Line	40 (4.25%)	100.00%	0.90 (SD 1.83)	6.90%
Kicker	0 (0.00%)	N/A	N/A	N/A
Linebacker	30 (3.50%)	100.00%	2.75 (SD 3.98)	20.83%
Long Snapper	3 (4.92%)	100.00%	8.00 (N/A)	100.00%
Offensive Line	27 (3.66%)	98.30%	3.04 (SD 2.52)	18.52%
Punter	1 (1.08%)	100.00%	1.00 (N/A)	0.00%
Quarterback	14 (6.67%)	93.33%	1.04 (SD 1.98)	26.67%
Running Back	26 (4.46%)	96.30%	1.27 (SD 1.93)	14.81%
Tight End	16 (4.19%)	100.00%	2.50 (SD 4.55)	12.50%
Wide Receiver	39 (5.03%)	100.00%	1.13 (SD 1.91)	5.00%

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Table 6. Finger Injury Outcomes Stratified by Player Position.				
Position	**Number of Players with Finger Injuries**	**% Return to Play**	**Number of Games Missed**	**% Injured Reserve**
Total	149 (2.43%)	98.68%	0.72 (SD 2.04)	4.61%
Center	7 (3.93%)	100.00%	0.17 (SD 0.41)	0.00%
Defensive Back	22 (1.82%)	100.00%	1.00 (SD 1.85)	9.09%
Defensive Line	13 (1.38%)	100.00%	0.13 (SD 0.35)	0.00%
Kicker	0 (0.00%)	N/A	N/A	N/A
Linebacker	16 (1.86%)	93.33%	0.36 (SD 0.50)	6.67%
Long Snapper	1 (1.64%)	100.00%	0.00 (N/A)	0.00%
Offensive Line	21 (2.85%)	94.74%	3.45 (SD 1.44)	10.53%
Punter	0 (0.00%)	N/A	N/A	N/A
Quarterback	16 (7.62%)	100.00%	0.89 (SD 1.45)	5.56%
Running Back	11 (1.89%)	100.00%	0.00 (SD 0.00)	0.00%
Tight End	12 (3.14%)	100.00%	0.45 (SD 0.69)	0.00%
Wide Receiver	30 (3.87%)	100.00%	0.91 (SD 2.86)	2.94%

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Table 7. Thumb Injury Outcomes Stratified by Player Position.				
Position	**Number of Players with Thumb Injuries**	**% Return to Play**	**Number of Games Missed**	**% Injured Reserve**
Total	149 (2.43%)	98.45%	1.97 (SD 3.76)	15.46%
Center	7 (3.93%)	83.33%	0.80 (SD 1.79)	33.33%
Defensive Back	42 (3.47%)	100.00%	1.95 (SD 3.73)	13.64%
Defensive Line	19 (2.02%)	100.00%	1.08 (SD 2.45)	12.50%
Kicker	0 (0.00%)	N/A	N/A	N/A
Linebacker	27 (3.15%)	100.00%	0.96 (SD 1.75)	3.57%
Long Snapper	0 (0.00%)	N/A	N/A	N/A
Offensive Line	22 (2.99%)	100.00%	4.15 (SD 2.08)	16.67%
Punter	0 (0.00%)	N/A	N/A	N/A
Quarterback	23 (10.95%)	96.43%	4.74 (SD 3.41)	25.00%
Running Back	14 (2.40%)	93.33%	1.07 (SD 1.98)	13.33%
Tight End	4 (1.05%)	100.00%	0.00 (SD 0.00)	0.00%
Wide Receiver	20 (2.58%)	100.00%	3.80 (SD 5.66)	25.00%

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